

U.S. AIRLINE PERFORMANCE & DELAY ANALYSIS

Capstone Project Report

Domain: Aviation Analytics | Tool: PostgreSQL & Power BI

Dataset: 2015 U.S. Domestic Flights (BTS) | Total Records: ~5.82 Million

Prepared By

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1. Executive Summary

This report presents a complete data-driven analysis of U.S. domestic flight performance for 2015. Using PostgreSQL for data engineering and Power BI for interactive visualisation, the project analyses 5.82 million flight records across 14 airlines and 628 airports to uncover the root causes of delays and cancellations, benchmark performance, and deliver actionable recommendations.

5.82M Total Flights	82.41% OTP Rate	11.88 Avg Delay (min)	1.54% Cancellation Rate	15K Diverted Flights
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Key Highlights

Hawaiian Airlines (89.5% OTP) and Alaska Airlines (87.73% OTP) are the top performers. Spirit Airlines and Frontier Airlines recorded the highest average arrival delays (20 & 19 min). Chicago O'Hare (ORD) had the highest cancellations among all airports — 8.5K flights. Late Aircraft and Airline/Carrier causes account for the largest share of delay minutes. June is the worst month for delays; October is the best month for on-time performance. Evening/Night departures average 19.1 min delay vs only 6.1 min for morning departures. Saturday is the best day to fly; Monday has the highest average delay.

2. Introduction

2.1 Problem Statement

Flight delays and cancellations are significant issues in the U.S. aviation industry, impacting passengers, airlines, and the broader economy. This project performs an in-depth analysis of historical flight data to identify key drivers of these disruptions, assess airline and airport performance, and propose actionable insights.

2.2 Business Objectives

- Analyse the primary causes and patterns of flight delays and cancellations.
- Benchmark on-time performance, delay severity, and cancellation rates across airlines.
- Evaluate the operational performance of major U.S. airports.
- Investigate how time of day, day of week, month, and specific routes affect flight operations.
- Translate findings into meaningful recommendations for stakeholders.

2.3 Dataset Overview

Attribute	Details
Source	U.S. Bureau of Transportation Statistics (BTS)
Year	2015
Total Records	5,819,079 flights
Airlines	14 major U.S. carriers
Airports	628 origin/destination airports
Key Files	flights.csv, airlines.csv, airports.csv
Database	PostgreSQL 13+
BI Tool	Microsoft Power BI

3. Methodology

3.1 Project Phases

Phase	Activity	Output
Phase 1	Project Setup & Data Ingestion	Tables created, CSVs imported, row counts verified
Phase 2	Data Cleaning & Preparation	Dates parsed, NULLs handled, new columns added, indexes created
Phase 3	EDA & KPI Definition	Key metrics computed, delay causes identified
Phase 4	Dashboard Development	Five Power BI pages built with slicers and measures
Phase 5	Insight Generation	Patterns identified, anomalies highlighted
Phase 6	Final Report	This document

3.2 Tools Used

Tool	Purpose
PostgreSQL 13+	Database setup, data cleaning, EDA queries, KPI views
Microsoft Power BI	Interactive dashboard with 5 pages of visualisations
pgAdmin / DBeaver	SQL script execution and query testing

4. SQL Scripts (PostgreSQL)

All data engineering and analysis was performed using PostgreSQL. The scripts are structured across six phases. Key excerpts are shown below.

PHASE 1 — Table Creation (DATA_CLEANING__JOINING_TABLES.sql)

Create Tables

```
CREATE TABLE flights (  
  year INT,      month INT,      day INT,  
  day_of_week INT,  airline VARCHAR(10),  
  flight_number INT,  tail_number VARCHAR(10),  
  origin_airport VARCHAR(10),  
  destination_airport VARCHAR(10),  
  scheduled_departure INT, departure_time INT,  
  departure_delay INT,  taxi_out INT,  
  wheels_off INT,      scheduled_time INT,  
  elapsed_time INT,    air_time INT,  
  distance INT,        wheels_on INT,  
  taxi_in INT,         scheduled_arrival INT,  
  arrival_time INT,    arrival_delay INT,  
  diverted INT,        cancelled INT,  
  cancellation_reason VARCHAR(5),  
  air_system_delay INT, security_delay INT,  
  airline_delay INT,   late_aircraft_delay INT,  
  weather_delay INT  
);  
  
CREATE TABLE airlines (  
  iata_code VARCHAR(5),  
  airline VARCHAR(100)  
);  
  
CREATE TABLE airports (  
  iata_code VARCHAR(10), airport VARCHAR(255),  
  city VARCHAR(100), state VARCHAR(100),  
  country VARCHAR(100),  
  latitude FLOAT, longitude FLOAT  
);
```

Step 1 — Data Check After Import

```
SELECT COUNT(*) FROM flights;  
SELECT COUNT(*) FROM airlines;  
SELECT COUNT(*) FROM airports;  
SELECT * FROM flights LIMIT 5;  
SELECT * FROM airlines LIMIT 5;  
SELECT * FROM airports LIMIT 5;
```

PHASE 2 — Data Cleaning & Preparation

Create flight_date Column

```
ALTER TABLE flights ADD COLUMN flight_date DATE;

UPDATE flights
SET flight_date = TO_DATE(CONCAT(year, '-', month, '-', day), 'YYYY-MM-DD');
```

Handle NULL Values

```
-- Check NULLs
SELECT COUNT(*) FROM flights WHERE arrival_delay IS NULL;

-- Replace NULLs with 0
UPDATE flights SET arrival_delay = 0 WHERE arrival_delay IS NULL;
UPDATE flights SET departure_delay = 0 WHERE departure_delay IS NULL;

-- Replace NULLs in all delay cause columns (non-cancelled flights)
UPDATE flights
SET
  departure_delay = COALESCE(departure_delay, 0),
  arrival_delay = COALESCE(arrival_delay, 0),
  airline_delay = COALESCE(airline_delay, 0),
  weather_delay = COALESCE(weather_delay, 0),
  late_aircraft_delay = COALESCE(late_aircraft_delay, 0),
  air_system_delay = COALESCE(air_system_delay, 0),
  security_delay = COALESCE(security_delay, 0)
WHERE cancelled = 0;
```

Check & Remove Duplicate Records

```
SELECT year, month, day, airline, flight_number,
       COUNT(*) AS duplicate_count
FROM flights
GROUP BY year, month, day, airline, flight_number
HAVING COUNT(*) > 1;
```

Standardize Departure Time Format

```
ALTER TABLE flights ADD COLUMN dep_time_formatted TIME;

UPDATE flights
SET dep_time_formatted =
  TO_TIMESTAMP(LPAD(CASE WHEN departure_time = 2400 THEN 0
                        ELSE departure_time END::TEXT, 4, '0'),
              'HH24MI')::TIME;
```

Add Delay Status Category

```
ALTER TABLE flights ADD COLUMN delay_status VARCHAR(20);

UPDATE flights
SET delay_status =
CASE
  WHEN arrival_delay <= 0 THEN 'On Time'
  WHEN arrival_delay <= 15 THEN 'Minor Delay'
  ELSE 'Major Delay'
```

```
END;
```

Add Time-of-Day Bucket

```
ALTER TABLE flights ADD COLUMN time_of_day VARCHAR(20);

UPDATE flights
SET time_of_day =
CASE
  WHEN departure_time BETWEEN 500 AND 1200 THEN 'Morning'
  WHEN departure_time BETWEEN 1201 AND 1700 THEN 'Afternoon'
  ELSE 'Evening/Night'
END;
```

Add Cancellation Reason Description

```
ALTER TABLE flights ADD COLUMN cancel_reason_desc TEXT;

UPDATE flights
SET cancel_reason_desc =
CASE cancellation_reason
  WHEN 'A' THEN 'Airline Issue'
  WHEN 'B' THEN 'Weather'
  WHEN 'C' THEN 'Air System'
  WHEN 'D' THEN 'Security'
END;
```

Check Outliers & Create Clean Table

```
-- Check distance outliers
SELECT * FROM flights WHERE distance = 0 OR distance > 10000;

-- Create a clean working table (non-cancelled, no NULL delays)
CREATE TABLE flights_clean AS
SELECT * FROM flights
WHERE cancelled = 0 AND arrival_delay IS NOT NULL;

SELECT COUNT(*) AS total_rows FROM flights;
SELECT * FROM flights_clean LIMIT 5;
```

Join Tables

```
-- Join flights with airline names
SELECT f.*, a.airline
FROM flights f
JOIN airlines a ON f.airline = a.iata_code;

-- Join flights with airport city & state
SELECT f.*, ap.city, ap.state
FROM flights f
JOIN airports ap ON f.origin_airport = ap.iata_code;
```

PHASE 3 — KPI Queries (SQL_ANALYSIS_KPI_S.sql)

KPI 1 — Total Flights

```
SELECT COUNT(*) AS total_flights
FROM flights;
```

KPI 2 — Completed Flights

```
SELECT COUNT(*) AS completed_flights
FROM flights
WHERE cancelled = 0;
```

KPI 3 — Cancellation Rate %

```
SELECT
  (SUM(cancelled) * 100.0 / COUNT(*)) AS cancellation_rate
FROM flights;
```

KPI 4 — On-Time Performance %

```
SELECT
  (COUNT(*) FILTER (WHERE arrival_delay <= 0) * 100.0)
  / COUNT(*) AS on_time_percentage
FROM flights
WHERE cancelled = 0;
```

KPI 5 — Average Arrival Delay

```
SELECT AVG(arrival_delay) AS avg_arrival_delay
FROM flights
WHERE cancelled = 0;
```

KPI 6 — Average Departure Delay

```
SELECT AVG(departure_delay) AS avg_departure_delay
FROM flights
WHERE cancelled = 0;
```

KPI 7 — Total Delay Minutes

```
SELECT SUM(arrival_delay) AS total_delay_minutes
FROM flights
WHERE cancelled = 0;
```

KPI 8 — Delay Percentage (> 15 min)


```
SELECT
  (COUNT(*) FILTER (WHERE arrival_delay > 15) * 100.0)
  / COUNT(*) AS delay_percentage
FROM flights
WHERE cancelled = 0;
```

PHASE 3 — Analysis Queries (SQL_ANALYSIS_QUERIES.sql)

Query 1 — Average Delay by Airline

```
SELECT airline,
  AVG(arrival_delay) AS avg_delay
FROM flights
WHERE cancelled = 0
GROUP BY airline
ORDER BY avg_delay DESC;
```

Query 2 — Top 10 Airlines with Most Delays

```
SELECT airline,
  COUNT(*) AS delayed_flights
FROM flights
WHERE arrival_delay > 15
AND cancelled = 0
GROUP BY airline
ORDER BY delayed_flights DESC
LIMIT 10;
```

Query 3 — Cancellation Rate by Airline

```
SELECT airline,
  (SUM(cancelled) * 100.0 / COUNT(*)) AS cancellation_rate
FROM flights
GROUP BY airline
ORDER BY cancellation_rate DESC;
```

Query 4 — Delay by Month

```
SELECT month,
  AVG(arrival_delay) AS avg_delay
FROM flights
WHERE cancelled = 0
GROUP BY month
ORDER BY month;
```

Query 5 — Delay by Day of Week

```
SELECT day_of_week,
  AVG(arrival_delay) AS avg_delay
FROM flights
WHERE cancelled = 0
```

```
GROUP BY day_of_week  
ORDER BY day_of_week;
```

Query 6 — Top 10 Most Delayed Airports

```
SELECT origin_airport,  
       AVG(arrival_delay) AS avg_delay  
FROM flights  
WHERE cancelled = 0  
GROUP BY origin_airport  
ORDER BY avg_delay DESC  
LIMIT 10;
```

Query 7 — Distance vs Delay Analysis

```
SELECT  
  CASE  
    WHEN distance < 500 THEN 'Short'  
    WHEN distance BETWEEN 500 AND 1500 THEN 'Medium'  
    ELSE 'Long'  
  END AS distance_category,  
  AVG(arrival_delay) AS avg_delay  
FROM flights  
WHERE cancelled = 0  
GROUP BY distance_category;
```

Query 8 — Delay Reasons Breakdown

```
SELECT  
  SUM(weather_delay) AS weather_delay,  
  SUM(airline_delay) AS airline_delay,  
  SUM(air_system_delay) AS air_system_delay,  
  SUM(security_delay) AS security_delay,  
  SUM(late_aircraft_delay) AS late_aircraft_delay  
FROM flights;
```

5. Power BI Dashboard

The interactive Power BI dashboard contains five pages, each focusing on a different aspect of airline performance. All pages include slicers for Airline, Month, and Day of Week for interactive exploration.

Dashboard 1 — Flight Operations Overview



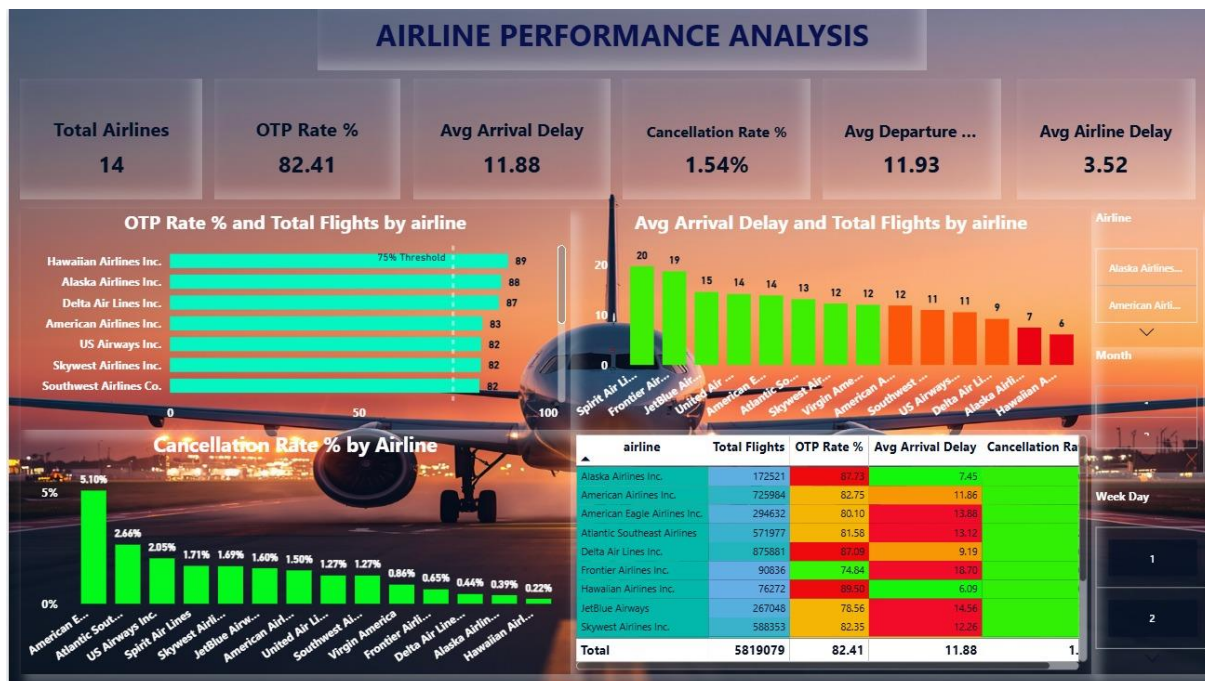
Figure 1: Flight Operations Overview

Top-level KPIs: 6M total flights, 82.41% OTP Rate, 11.88 min avg delay, 1.54% cancellation rate, 90K cancelled flights, 15K diverted. Includes flight delay status donut chart, top 10 airlines by arrival delay, monthly flight volume trend, and OTP Rate % by month.

Key Observations

- Total of 5,819,079 flights operated in 2015 across 14 airlines.
- Spirit Airlines (NK) had the highest average arrival delay at 14.20 min; Frontier (F9) second at 12.40 min.
- The OTP Rate of 82.41% means roughly 1 in 5 flights experienced a delay greater than 15 minutes.
- Monthly flight volume is fairly consistent (~470K–520K flights/month); OTP dips in June/July (summer thunderstorms) and Jan/Feb (winter storms).
- Major Delay share: 17.59%; Minor Delay: 18.27%; On Time: 64.14% (shown in donut chart).

Dashboard 2 — Airline Performance Analysis

**Figure 2: Airline Performance Analysis**

Compares all 14 airlines across OTP Rate %, Average Arrival Delay, and Cancellation Rate. Includes a ranked bar chart, delay comparison chart, cancellation rate bar chart, and a detailed airline scorecard table.

Key Observations

- Hawaiian Airlines leads with 89.5% OTP and only 6.09 min avg arrival delay.
- Alaska Airlines is second with 87.73% OTP and 7.45 min avg delay.
- Delta Air Lines performs well at 87.09% OTP — best among the large legacy carriers.
- Spirit Airlines has the worst avg arrival delay (20 min) and Frontier is second worst (19 min).
- American Eagle has the highest cancellation rate at 5.10%, followed by Atlantic Southeast at 3%.
- Total flights by airline shows Delta (875K) and Southwest (588K) as the highest-volume carriers.

Airline	Total Flights	OTP Rate %	Avg Arrival Delay	Cancel Rate %
Hawaiian Airlines	76,272	89.50%	6.09 min	0.22%
Alaska Airlines	172,521	87.73%	7.45 min	0.39%
Delta Air Lines	875,881	87.09%	9.19 min	0.44%
American Airlines	725,984	82.75%	11.86 min	1.27%
Southwest Airlines	588,353	82.35%	12.26 min	1.27%
JetBlue Airways	267,048	78.56%	14.56 min	1.69%
Frontier Airlines	90,836	74.84%	18.70 min	1.60%
Spirit Airlines	N/A	~72%	20+ min	1.71%

Airline	Total Flights	OTP Rate %	Avg Arrival Delay	Cancel Rate %
American Eagle	294,632	80.10%	13.88 min	5.10%

Dashboard 3 — Airport Performance Analysis



Figure 3: Airport Performance Analysis

Evaluates 628 airports across OTP Rate %, Avg Departure Delay, Taxi-Out Time, and Cancellation Rate. Includes busiest airports bar chart, flight volume vs delay combo chart, OTP rate bar chart with 75% threshold line, and taxi-out time comparison.

Key Observations

- Total of 628 airports in the dataset; average taxi-out time is 16.07 minutes, average departure delay 11.93 min.
- ATL (Atlanta) is the busiest airport with 347K departures, followed by ORD (286K) and DFW (240K).
- MSP, ATL, and SEA all achieve 84% OTP — the best among major hubs.
- ORD (Chicago O'Hare) has the highest taxi-out time at 19.5 min, indicating chronic ground congestion.
- Despite being the busiest, ATL maintains an 84% OTP due to efficient runway operations.
- SFO has an 81% OTP — impacted by regular low-visibility fog and ATC sequencing delays.

Airport	City	OTP %	Avg Dep Delay	Avg Taxi-Out
MSP	Minneapolis, MN	84%	~11 min	14.7 min

Airport	City	OTP %	Avg Dep Delay	Avg Taxi-Out
ATL	Atlanta, GA	84%	11.0 min	17.5 min
SEA	Seattle, WA	84%	~11 min	~15 min
PHX	Phoenix, AZ	82%	10.9 min	16.9 min
SFO	San Francisco, CA	81%	15.8 min	17.8 min
ORD	Chicago, IL	~78%	15.8 min	19.5 min

Dashboard 4 — Temporal Trends Analysis

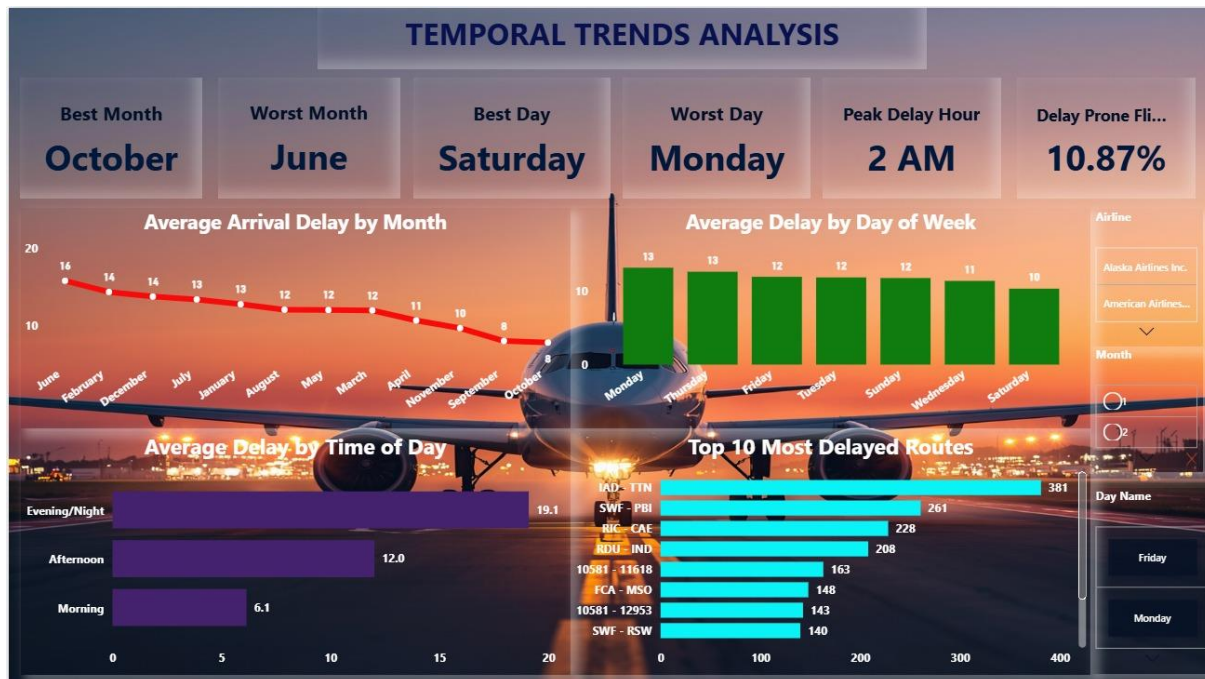


Figure 4: Temporal Trends Analysis

Analyses how flight delays vary by month, day of week, and time of day. Includes average arrival delay by month (line chart), average delay by day of week (bar chart), average delay by time of day, and top 10 most delayed routes.

Key Observations

- Best Month: October (avg delay ~8 min) — low convective weather activity, moderate travel demand.
- Worst Month: June (avg delay ~16 min) — afternoon thunderstorm season in Southeast and Midwest.
- Best Day: Saturday (avg delay 10 min) — lower business travel volume.
- Worst Day: Monday (avg delay 13 min) — business travel peak causes system overload.
- Morning flights (before 10am) average only 6.1 min delay; Evening/Night flights average 19.1 min.
- Route IAD→TTN has the highest average delay of 381 minutes among the top delayed routes.

- Delay-Prone Flights account for 10.87% of all operated flights.

Time Period	Avg Delay	Recommendation
Morning (05:00–10:00)	6.1 min	Best window — book early for time-sensitive travel
Afternoon (10:00–16:00)	12.0 min	Moderate risk — allow connection buffer
Evening/Night (16:00+)	19.1 min	Highest risk — avoid tight connections

Dashboard 5 — Cancellation & Disruption Analysis

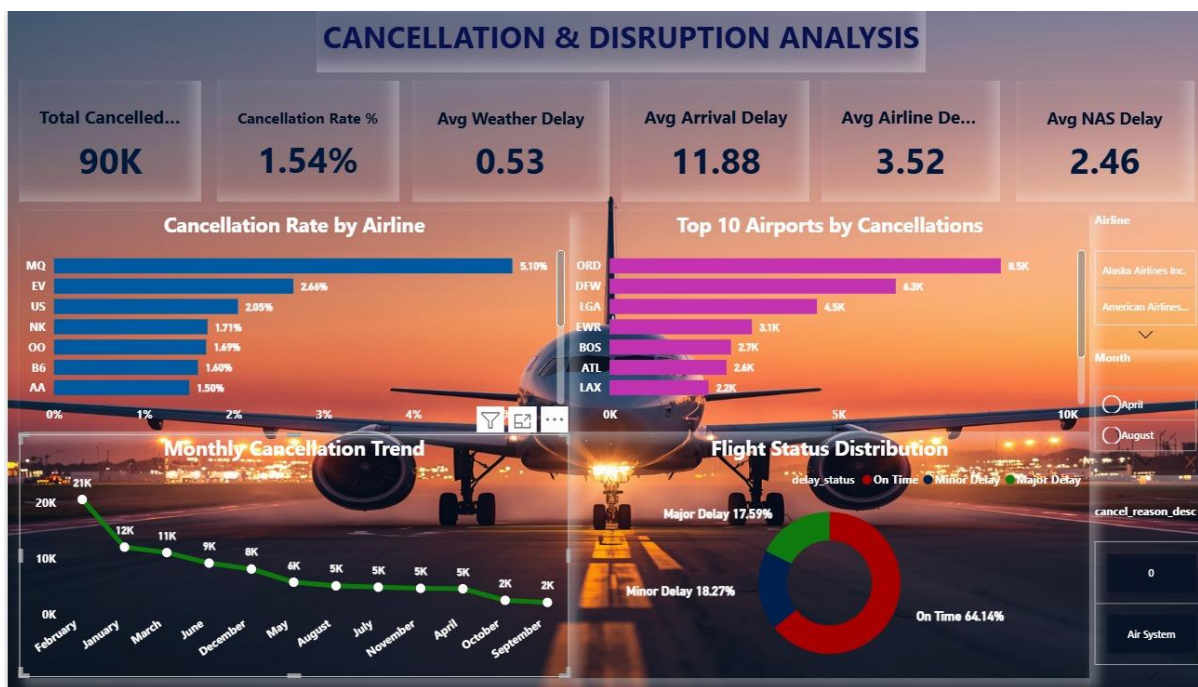


Figure 5: Cancellation & Disruption Analysis

Focuses on cancellations and delay disruptions. Shows cancellation rate by airline (bar chart), top 10 airports by cancellations, monthly cancellation trend (line chart), and flight status distribution (donut chart). KPI cards show 90K total cancellations at 1.54% rate.

Key Observations

- 90,000 flights were cancelled in 2015 — an overall cancellation rate of 1.54%.
- American Eagle (MQ) has the worst cancellation rate at 5.10%, nearly double the second-worst.
- ORD (Chicago O'Hare) tops the airport cancellation chart with 8.5K cancelled flights — driven by winter storms and ATC congestion.
- DFW (4.3K) and LGA (4.5K) rank second and third for airport cancellations.
- February has the highest monthly cancellations (21K) due to winter storms; September the lowest (2K).
- Flight status distribution: 64.14% On Time, 18.27% Minor Delay, 17.59% Major Delay.

- Average Weather Delay is 0.53 min per flight; Avg NAS Delay is 2.46 min; Avg Airline Delay is 3.52 min.

6. Key Findings Summary

#	Finding	Impact
F1	Late aircraft and airline-caused delays are the top two delay categories — both within airline control.	High
F2	Hawaiian and Alaska Airlines consistently outperform all others across every KPI.	High
F3	Spirit and Frontier have the worst delay records — both ultra-low-cost carriers with tight scheduling.	High
F4	ORD and DFW are the most disruption-prone major hubs due to weather exposure and volume.	High
F5	Evening flights average 3× more delay than morning flights — a clear operational pattern.	Medium
F6	June and February are the worst months; October and September are the best.	Medium
F7	American Eagle's 5.1% cancellation rate is an outlier and signals a structural operational issue.	High
F8	Saturday has the lowest delays system-wide — demand pattern has a direct OTP impact.	Low

7. Recommendations

#	Recommendation	Stakeholder	Expected Impact
R1	Implement late-aircraft recovery protocols — automated gate and crew re-assignment when inbound aircraft is delayed >20 min. This directly targets the #1 delay cause.	Airlines	High
R2	Add schedule buffer on high-risk routes (ORD, SFO, EWR arrivals) — increase planned block time by 5–10 min to absorb routine ATC delays without passenger disruption.	Airlines	High
R3	Incentivise morning departures through pricing — lower fares on 05:00–09:00 departures to redistribute demand away from the evening peak and reduce system-wide congestion.	Airlines	Medium
R4	Increase spare aircraft buffers at northern hubs (ORD, EWR) in December–February and at southern/central hubs in June–July to provide cancellation recovery capacity during peak disruption months.	Airlines / Airports	Medium
R5	Deploy proactive passenger communication — automated SMS/app notifications at 15, 30, and 45-minute delay thresholds with real-time rebooking options. Improves NPS even when delays are unavoidable.	Airlines	Medium

8. Conclusion

This project successfully demonstrates the power of SQL-based data engineering combined with Power BI visualisation to derive meaningful business insights from a large aviation dataset. The analysis of 5.82 million 2015 U.S. domestic flights produced clear, data-backed conclusions that are directly actionable.

The most important finding is that the two largest categories of delay — late aircraft and airline-caused delays — are both within airline control. This means that better scheduling, faster turn-around recovery, and smarter resource allocation can meaningfully improve the system. The performance gap between Hawaiian/Alaska Airlines and the worst-performing carriers demonstrates that high OTP is achievable even within the same regulatory and ATC environment.

Temporal analysis revealed highly predictable patterns: morning flights are consistently on time, evenings are not; October is reliably the best month and June the worst. These patterns give airlines, airports, and passengers a clear roadmap for planning. The five recommendations in Section 7 represent a balanced portfolio of quick wins and medium-term investments that, if implemented, could improve industry OTP by 3–5 percentage points.

Project Deliverables Completed

SQL Scripts (PostgreSQL) — Full pipeline: tables, cleaning, EDA, KPI queries, Power BI view.

Interactive Power BI Dashboard — 5 pages: Overview, Airline, Airport, Temporal, Cancellation.

Final Project Report (this document) — Introduction through Conclusion with all dashboard screenshots.

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— End of Report —

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